Thermodynamics Formula Sheet

Ch 6: The Second Law of Thermodynamics (Entropy)

Units and Conversions

SI Prefixes

femto (f)	$= 10^{-15}$	deca (da)	$= 10^{1}$
pico (p)	$= 10^{-12}$	hecto (h)	$= 10^{2}$
nano (n)	$= 10^{-9}$	kilo (k)	$= 10^{3}$
micro (μ)	$= 10^{-6}$	mega (M)	$= 10^{6}$
milli (m)	$= 10^{-3}$	giga (G)	$= 10^{9}$
centi (c)	$= 10^{-2}$	tera (T)	$= 10^{12}$
deci (d)	$= 10^{-1}$	peta (P)	$= 10^{15}$

Temperature Conversions

- Absolute temperature conversions
  - $T_K = T_{\circ C} + 273.15$
  - $T_{\circ R} = T_{\circ F} + 459.67$
  - $T_{\circ F} = \frac{9}{5} T_{\circ C} + 32$
  - $T_{\circ R} = \frac{9}{5} T_K$
- Temperature difference conversions
  - $\Delta K = \Delta ^\circ C$
  - $\Delta ^\circ R = \Delta ^\circ F$
  - $\Delta ^\circ F = \frac{9}{5} \Delta ^\circ C$
  - $\Delta ^\circ R = \frac{9}{5} \Delta K$

Density and Specific Gravity

$$\rho = \frac{m}{V\!\!\!/} \quad \text{(density)}, \qquad v\!\!\!/ = \frac{1}{\rho} = \frac{V\!\!\!/}{m} \quad \text{(specific volume)}$$

Specific gravity is defined as a relative density compared to water (at 4°C) where $\rho_{water} = 1000\,\mathrm{kg/m^3}$:

$$\mathrm{SG} = \frac{\rho_{fluid}}{\rho_{water}} \implies \rho_{fluid} = SG \cdot \rho_{water}$$

Specific Weight

$$\gamma_s = \rho g \quad \text{(specific weight)}$$

Where hydrostatic pressure can be written as $P = \gamma_s h$

Volume

$1\,\mathrm{m^3} = \underline{1000\,\mathrm{L}} = 61024\,\mathrm{in^3} = 35.315\,\mathrm{ft^3} = \underline{264.17\,\mathrm{gal}}$

Slugs and lbm

$$\left[\frac{1\,\mathrm{lbf}}{32.174\,\frac{\mathrm{lbm \cdot ft}}{\mathrm{s^2}}}\right], \quad \left[\frac{1\,\mathrm{slug}}{32.174\,\mathrm{lb},}\right], \quad \left[\frac{1\,\mathrm{lbf}}{1\,\frac{\mathrm{slug \cdot ft}}{\mathrm{s^2}}}\right]$$

Energy Units

1 Btu = 778.169 ft·lbf = 1055.06 J

Work Units

1 hp = 550 ft·lbf/s = 745.7 W

Distance

1 mile = 5280 ft

Ideal Gas Equation of State

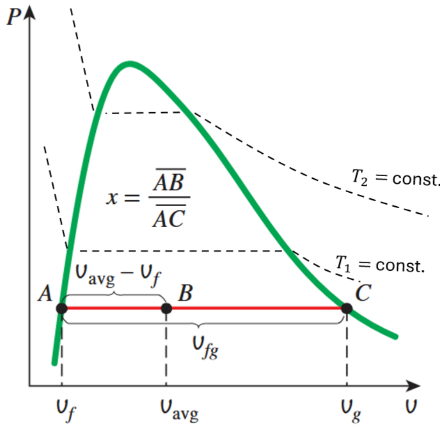
Refer to table A-4 for gas constants of common gases.

$$P v\!\!\!/ = RT, \quad R = \frac{\bar{R}}{M} \quad \left(\mathrm{kJ/kg \cdot K} \quad \text{or} \quad \frac{\mathrm{Btu/lbm \cdot R}}{\mathrm{psia \cdot ft^3/lbm \cdot R}}\right)$$

$$P V\!\!\!/ = mRT \quad \text{(in terms of TOTAL volume and mass)}$$

Universal gas constant: $\bar{R} = 8.314$ J/(mol·K)

Note: Convert units for P, $V\!\!\!/$, $v\!\!\!/$, and T to match the units of R when using the ideal gas law. Also, [kJ/kpa] = [m^3]



Phase Change & Saturation

$$h_{fg} = h_g - h_f \quad \text{(latent heat of vaporization)}$$

$$x = \frac{m_g}{m_f + m_g} \quad \text{(quality definition)}$$

$$\left.\begin{aligned} y_{avg} &= y_f + x(y_g - y_f) \quad (\text{for } v\!\!\!/, h, u) \\ x &= \frac{y_{avg} - y_f}{y_g - y_f} \quad \text{(solve for quality)} \end{aligned}\right\} y_{fg} = y_g - y_f$$

Compressed Liquid Approximation

$$y(T, P) \approx y_f(T) \quad \text{(use saturated liquid at same T)}$$

Pressure Relations

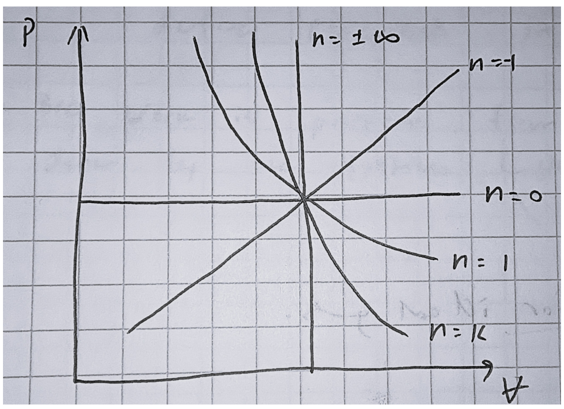
$$P = \frac{F}{A} = \rho g h \quad \text{(hydrostatic pressure)}$$

$$P_1 = P_2 \quad \Rightarrow \quad \frac{F_1}{A_1} = \frac{F_2}{A_2} \quad \text{(Pascal's Law)}$$

$$\Delta P = \rho g h \quad \text{(multi-fluid manometer)}$$

(add downward, subtract upward)

Polytropic Processes



Polytropic Processes (cont.)

$P\cancel{V}^n = \text{constant}$

- $n = 0$: isobaric process (constant pressure)
- $n = 1$: isothermal process (constant temperature, ideal gas)
- $n = k$: isentropic process (constant entropy)
- $n = \infty$: isochoric process (constant volume)

Efficiency

Heat engine

$$\eta_{HE} = \frac{W_{HE}}{Q_H} = \frac{Q_H - Q_L}{Q_H} \quad \leq \quad \eta_{Carnot,HE} = 1 - \frac{T_L}{T_H}$$

Heat Pump

$$\beta_{HP} = \frac{Q_H}{W_{HP}} = \frac{Q_H}{Q_H - Q_L} \quad \leq \quad \beta_{Carnot,HP} = \frac{T_H}{T_H - T_L}$$

Refrigerator

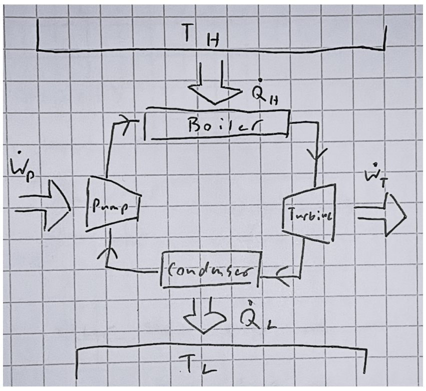
$$\beta_{Ref} = \frac{Q_L}{W_{Ref}} = \frac{Q_L}{Q_H - Q_L} \quad \leq \quad \beta_{Carnot,Ref} = \frac{T_L}{T_H - T_L}$$

Note: The heat and work terms can also be in rate forms $\dot{Q}$ and $\dot{W}$.

T_H and T_L are ABSOLUTE temperatures!!

Heat Engines

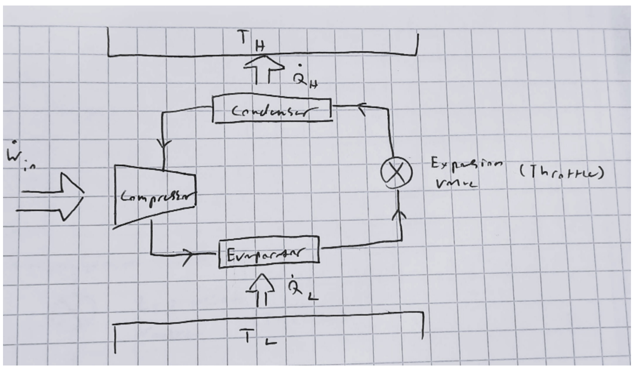
Kelvin-Planck: there has to exist losses as heat to surroundings (Q_L).
Produces work from energy input as heat (Q_H).



Note: $\dot{W}_{\text{net}} = \dot{W}_T - \dot{W}_P$

Refrigerators & Heat Pumps

Clausius: requires work input to transfer heat from cold to hot reservoir.
Refrigerators transfer heat from cold to hot reservoir, while heat pumps transfer heat from cold to hot reservoir. They have the same cycle but differ in *intent*.



First Law (Closed System)

$\Delta U = Q - W$ (energy balance for closed system)

$\dot{E}_{in} - \dot{E}_{out} = \frac{dE_{system}}{dt}$ (rate form)

First Law (Open System)

$$\frac{dE}{dt} = \dot{Q} - \dot{W} + \underbrace{\sum \dot{m}_i \left[h_i + \frac{1}{2}V_i^2 + gz_i\right]}_{\text{Inlet}} - \underbrace{\sum \dot{m}_e \left[h_e + \frac{1}{2}V_e^2 + gz_e\right]}_{\text{Outlet}}$$

For steady state process with no heat transfer:

$\dot{W} = \dot{m}e_{\text{mech}}$

$\dot{W} = \dot{m}\frac{\Delta P}{\rho} = \dot{\cancel{V}}\Delta P$ (pump power w/ no change in KE or PE)

Mass & Flow Relations

$\dot{m} = \rho\dot{\cancel{V}} = \rho A v_{avg}$ (mass flow rate)

$\dot{E} = \dot{m}e$ (energy flow rate)

Enthalpy

$h = u + P\cancel{v}, \quad \cancel{v} = \frac{1}{\rho}$ (specific enthalpy)

Work

$W = \int F \cdot dx, \quad \dot{W} = F \cdot v$ (general)

$$W = \int P\, d\cancel{V} \implies W = \frac{P_2\cancel{V}_2 - P_1\cancel{V}_1}{-n+1} \quad \text{(polytropic, } n \neq 1)$$

$$\implies W = C \ln\left(\frac{\cancel{V}_2}{\cancel{V}_1}\right) \quad \text{(isothermal, } n = 1)$$

$$\implies W = P\Delta\cancel{V} \quad \text{(isobaric)}$$

Where C is $P_1\cancel{V}_1^n = P_2\cancel{V}_2^n = C$ for a polytropic process.

$W = \frac{1}{2}k(x_2^2 - x_1^2)$ (spring work)

Heat Transfer

$Q = \int \dot{Q}\, dt = \dot{Q}\Delta t$ (total heat transferred)

$\dot{Q} = kA\frac{\Delta T}{\Delta x}$ (conduction), $\dot{Q} = hA(T_s - T_\infty)$ (convection)

$\dot{Q} = \epsilon\sigma A(T_s^4 - T_\infty^4)$ (radiation)

Specific Heat Capacity

$c_{\cancel{v}} = \left(\frac{\partial u}{\partial T}\right)_{\cancel{v}}$ (fixed volume), $c_p = \left(\frac{\partial h}{\partial T}\right)_p$ (fixed pressure)

$\implies \Delta u = c_{\cancel{v}}\Delta T, \quad \Delta h = c_p\Delta T$ (constant $c_{\cancel{v}}$ or c_p)

$c_p = c_{\cancel{v}} + R$ (ideal gas), $c_p = c_{\cancel{v}}$ (incompressible substances)

$Q = mc\Delta T$ ($c_{\cancel{v}}$ or c_p depending on the process)

$k = c_p/c_{\cancel{v}}$ (specific heat ratio), $c_{\text{water}} = 4.18\,\frac{\text{kJ}}{\text{kg}\cdot\text{K}}$

Isentropic Relations for Ideal Gases ($P\cancel{V}^k = \text{constant}$)

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{k-1}{k}}, \quad \frac{T_2}{T_1} = \left(\frac{\cancel{V}_1}{\cancel{V}_2}\right)^{k-1}, \quad \frac{P_2}{P_1} = \left(\frac{\cancel{V}_1}{\cancel{V}_2}\right)^{k}$$